

FERDIANSYACH

Data Analyst & ML Developer (Python, SQL) | Web-GIS & Predictive Modeling

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PROFESSIONAL SUMMARY

Data Analyst & Machine Learning Developer with strong expertise in Python, SQL, and geospatial analysis. Engineered end-to-end data pipelines and predictive ML models (LSTM, XGBoost) achieving 92% accuracy on 50,000+ data points at Telkom Indonesia. Experienced in building coastal water quality monitoring systems using Google Earth Engine with Mann-Kendall statistical tests. Adept at transforming raw, time-series datasets into actionable business insights through interactive dashboards (Streamlit).

PROFESSIONAL EXPERIENCE

Business Intelligence & ML Intern

Apr 2025 – Sep 2025

Telkom Indonesia | Infranexia | Internship | Jakarta, ID

- Engineered a predictive AI model using LSTM and Gradient Boosting that achieved 92% accuracy in smart building energy consumption forecasting, directly supporting the techno-economic analysis division.
- Architected an end-to-end data pipeline from raw ingestion through preprocessing to model deployment (LSTM, XGBoost), processing 50,000+ data points and generating actionable insights for energy efficiency optimization.
- Developed a web-based internship management platform in collaboration with 3 cross-functional divisions, serving 100+ interns and reducing onboarding coordination overhead.

Fullstack Developer

Apr 2024 – Nov 2024

UNAS FEST | Universitas Nasional Festival | Student Organization | Jakarta, ID

- Developed 10+ responsive website components using TypeScript and Tailwind CSS, achieving a Lighthouse score of 90+ for performance and accessibility on the official festival portal.
- Built Next.js API Routes for attendee registration and database integration, and engineered a testing pipeline with Jest and React Testing Library that slashed production bug rate by 60%.

Research & Development Assistant

Jun 2024 – Mar 2025

HIMASI | Universitas Nasional | Student Organization | Jakarta, ID

- Orchestrated a national technology seminar as Chief Executive, coordinating 15 committee members from planning to execution and successfully delivering the event to 150+ attendees on schedule.
- Developed and managed the association's official WordPress website, increasing website traffic by 40% and driving member engagement through structured, SEO-optimized content.

Computer Laboratory Assistant

Sep 2024 – Sep 2025

Lab Data Monetize | Universitas Nasional | Student Organization | Jakarta, ID

- Maintained and troubleshooted 30+ laboratory computer units, sustaining 98% device availability during operational hours and minimizing session downtime for 200+ students per semester.
- Optimized software installation and system configuration workflows, supporting lab sessions for 200+ students per semester through standardized maintenance procedures.

EDUCATION

Universitas Nasional

2022 – 2026

Bachelor of Computer Science (S.Kom) | Information Systems

GPA: 3.77

Capstone Project: Cloud-Based Coastal Water Quality Monitoring: Spatiotemporal Trend Analysis and Unsupervised Spectral Clustering via Web-GIS

Relevant Coursework: Web Programming, Database Systems, Software Engineering, Artificial Intelligence, System Analysis & Design, Computer Networks

TECHNICAL SKILLS

Data & Analysis: Python, Pandas, NumPy, Matplotlib, Tableau, Streamlit, GIS, spatial analysis

Machine Learning: Scikit-learn, XGBoost, LSTM / Keras, TensorFlow, clustering, unsupervised learning

Backend & DB: Prisma, Node.js, Express.js, MongoDB, MySQL / SQL, Postman

Frontend: React, Next.js, TypeScript, Tailwind CSS, HTML / CSS

DevOps & Tools: Jest & RTL, GitHub Actions, Git & GitHub, Google Cloud, Docker, Jupyter Notebook

SELECTED PROJECTS

Smart Meter Analysis (GitHub)

Predictive AI models (LSTM & XGBoost) for energy anomaly detection with a real-time interactive dashboard.

Python, Machine Learning, Data Analysis, Streamlit, XGBoost, LSTM, Pandas

Coastal Water Quality Monitoring (GitHub)

Coastal water quality monitoring for Jakarta & Banten via Google Earth Engine (2019–2025) using Sentinel-2, Landsat-8, K-Means, and Mann-Kendall.

Python, Google Earth Engine, Sentinel-2, Landsat-8, Streamlit, K-Means, Mann-Kendall, Remote Sensing

Indosaji E-commerce (GitHub)

An end-to-end full-stack e-commerce platform based on MERN with payment gateway integration.

React, Node.js, Express.js, MongoDB, REST API, JWT Auth, Stripe

CERTIFICATIONS & LICENSES

Junior Web Developer | BNSP (Badan Nasional Sertifikasi Profesi) (Feb 2026)

IT Specialist | Python | Certiport (Pearson VUE) (Oct 2025)

Scrum Mastery | Telkom Indonesia (Jul 2025)

Internet of Things Fundamental | Telkom Indonesia (Apr 2025)

Statistics Using R for Data Science | DQLab (Feb 2023)